Autoregressive Next-Word Text Generation

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1 Problem Statement

Build and compare Simple RNN, Bidirectional RNN, LSTM, and GRU models for autoregressive (next-word) text generation on *The War of the Worlds*.

2 Dataset

Project Gutenberg eBook #36 (`https://www.gutenberg.org/files/36/36-0.txt`). After cleaning: 70k tokens, vocabulary $\approx$ 3.1k, split 80/10/10, context length 20.

3 Method

Perplexity is defined as $PPL = \exp(\mathcal{L})$, where $\mathcal{L}$ is mean token cross-entropy. Sampling uses $P(w) \propto \exp(z_w/T)$ with $T \in \{0.5, 1.0, 1.5\}$.

4 Results

Training metrics are saved in `results/metrics.json`. Representative values will be inserted after the experiment completes:

Model	Params	Test PPL	Test Acc	Time(s)
SimpleRNN	–	–	–	–
BiRNN	–	–	–	–
LSTM	–	–	–	–
GRU	–	–	–	–

5 Conclusion

See `results/` and the Jupyter notebook `text_generation.ipynb`.